Software Engineering I (02161)

Agile Software Development

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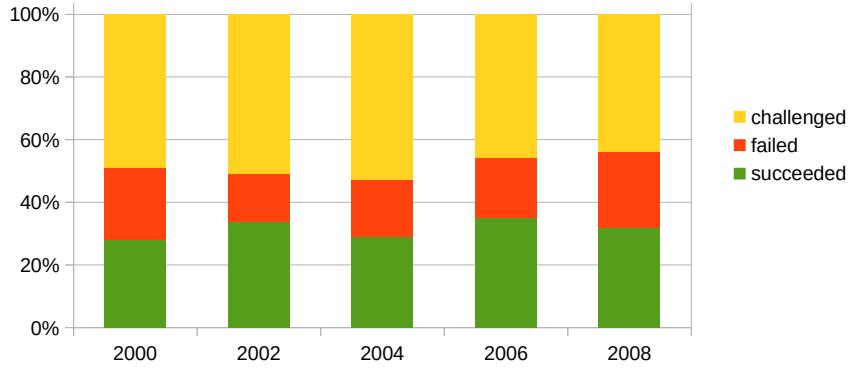
Success rate for software projects 2000—2008

- Challenges of Software Development
  - On **time**
  - In **budget**
  - No **defects**
  - Customer **satisfaction**



CHAOS Summary 2009 Report

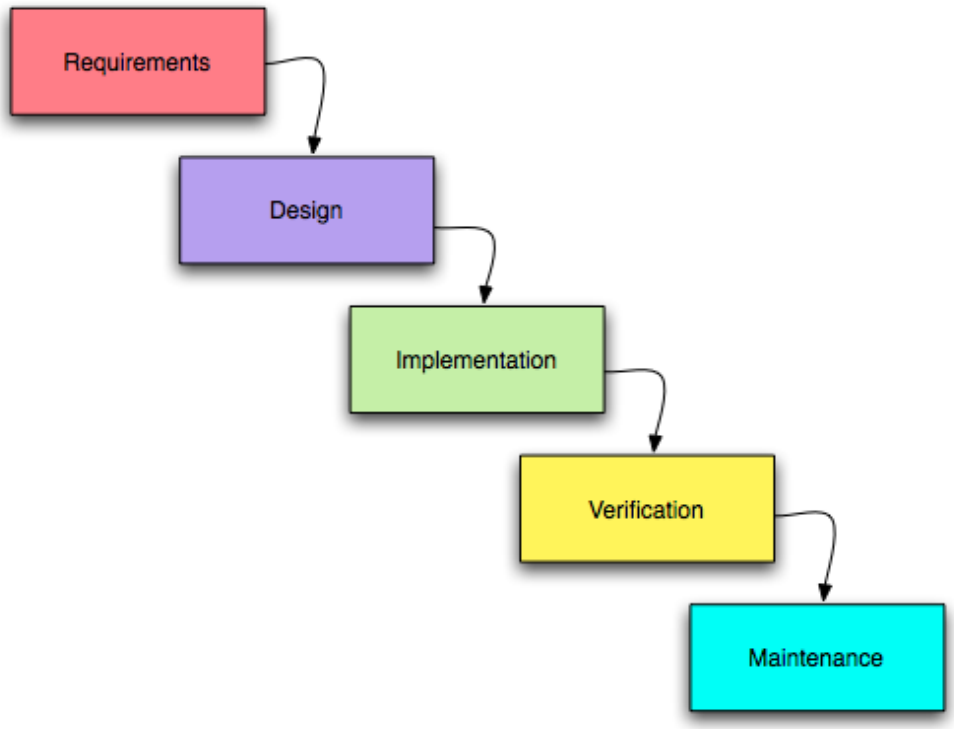
- Succeeded: 32%
- Failed: 20%
- Challenged: 48% (over time, over budget, . . .)

Activities in Software Development

- Understand and document what the customer wants: **Requirements Engineering**
- How to build the software: **Design**
- Build the software: **Implementation**
- Validate: **Testing**, **Verification**, **Evaluation**

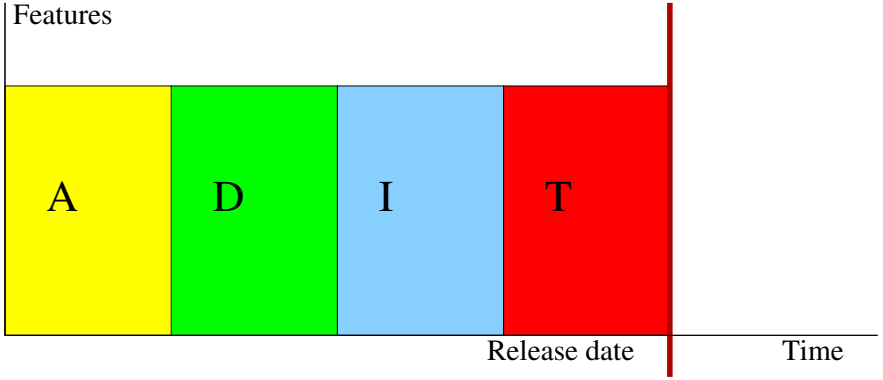
→ **Waterfall**

Waterfall process

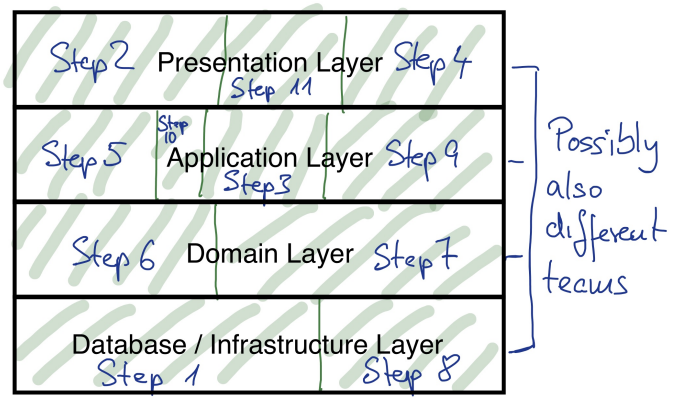


- 1970: Winston W. Royce how **not** to develop software
- 1985: Waterfall was required by the United States Department of Defence

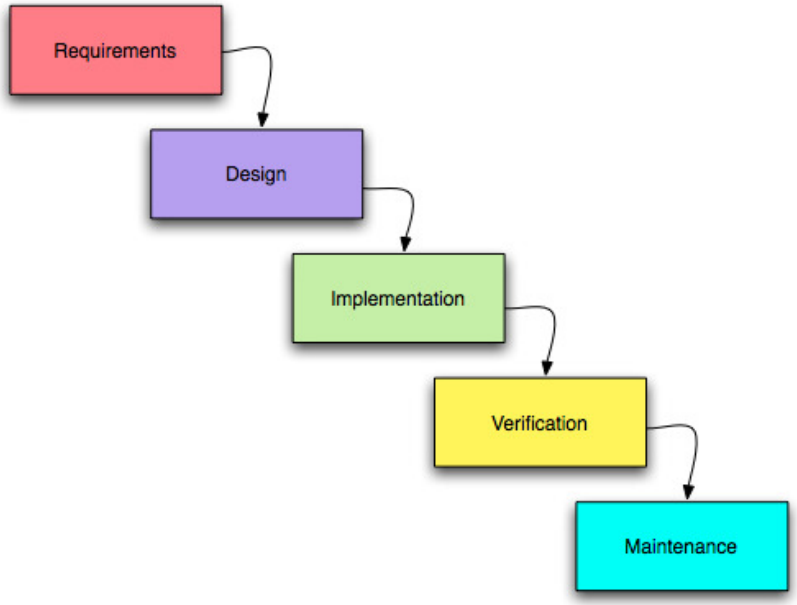
Delays in waterfall processes



Implementation by layers and not functionality



Costs of changing requirements: Waterfall

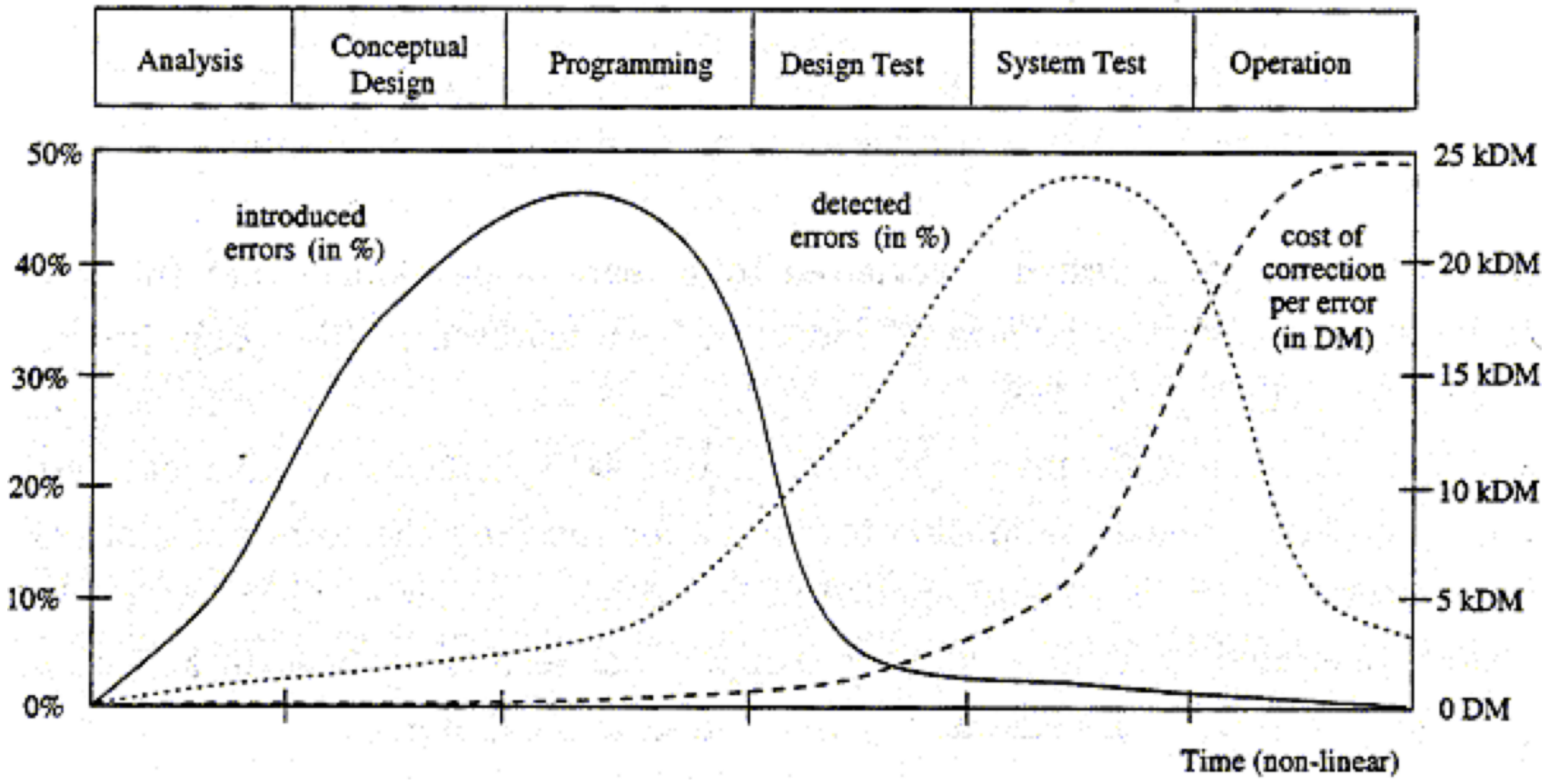


- Changed / new requirements change the design and implementation
  - Cost of change not predictable
  - → Avoid changing/new requirements if possible

→ Good for projects with simple precise requirements, like mathematical functions

→ Bad for projects with unclear requirements or with evolutionary requirements

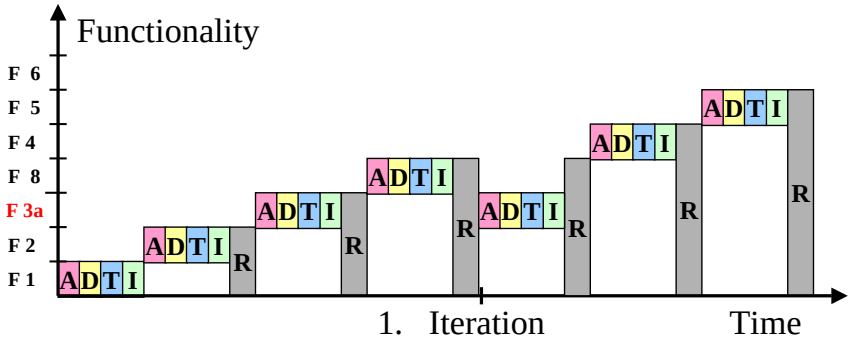
Problem in Software Engineering

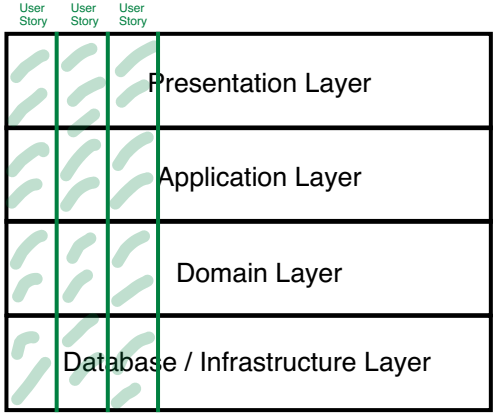


- Liggesmeyer 1998
- Improve the feedback cycle
  - Fail fast
  - Short Analysis, Design, Implement, Test cyles
  - One cycle per **user story**

Agile Software Development Methods (1999)

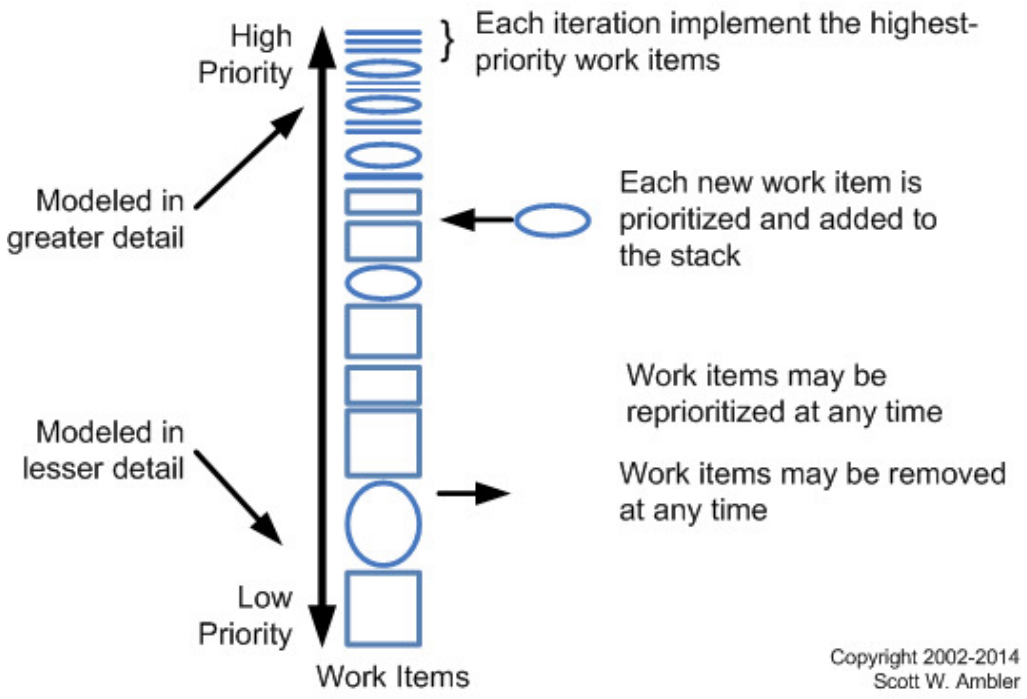
- Extreme Programming (XP) (1999), Scrum (1995–2001), Lean Software Development (2003), . . .
- Kanban (2010): not a method; *tool* to improve processes





- Highest priority user story first
- If delayed: important features are implemented

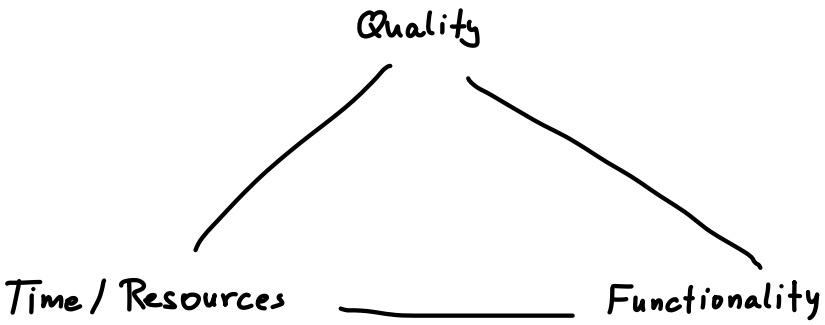
Changing Requirements: Agile Methods



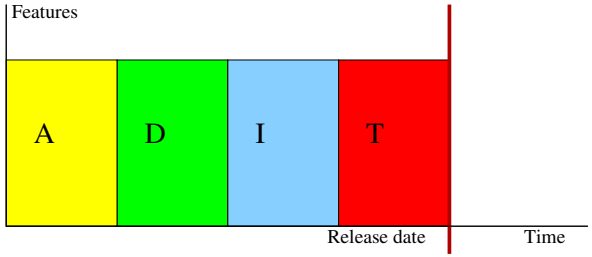
Scott Ambler 2003–2014 `http://www.agilemodeling.com/artifacts/userStory.htm`

- Cost of change
  - New / changed requirements not done yet: **zero costs**
  - Changed requirements already done: the cost of a requirement that will not be implemented

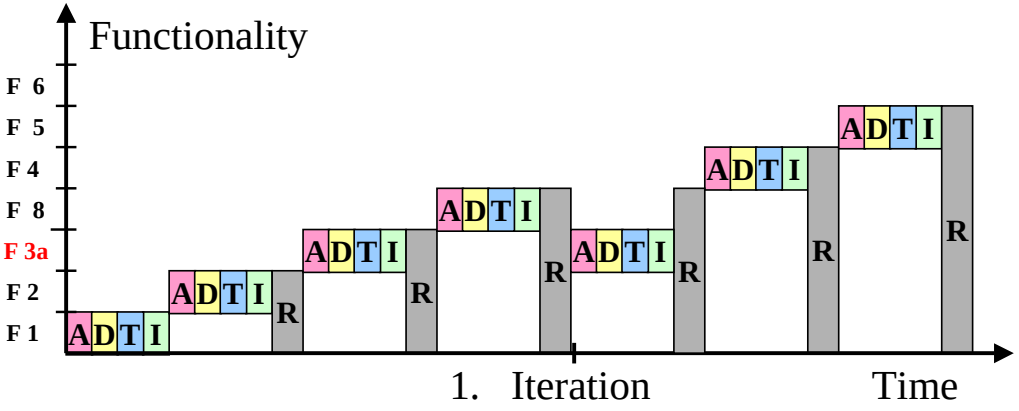
Resource Triangle



Waterfall

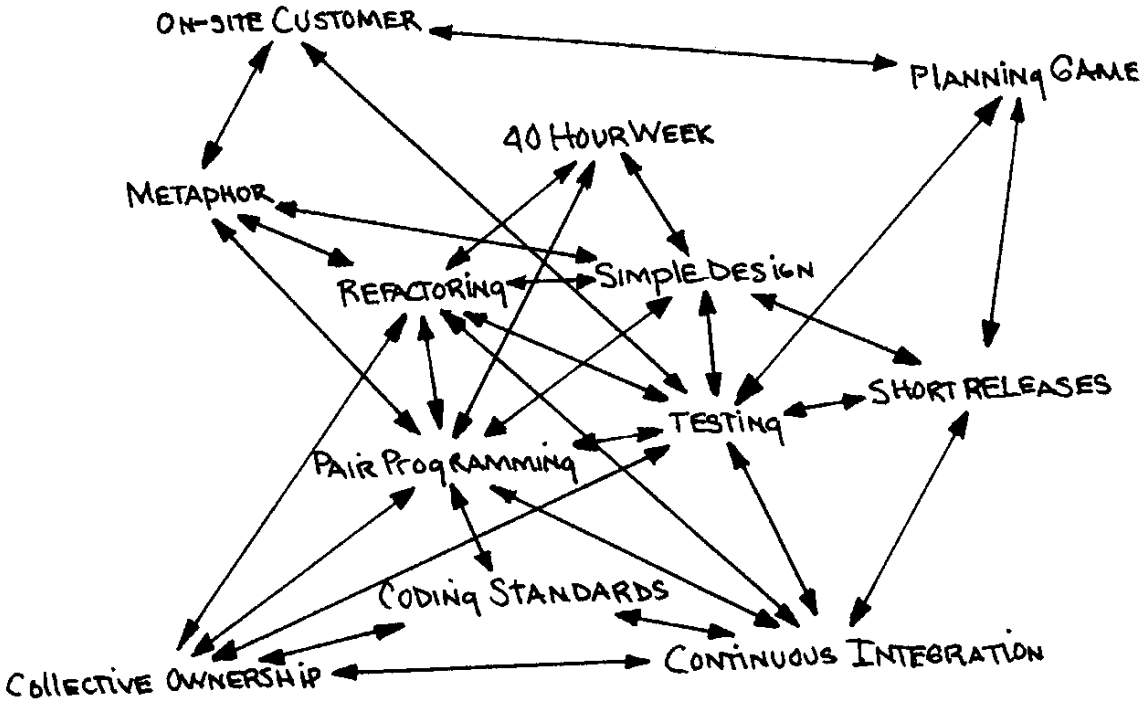


Agile



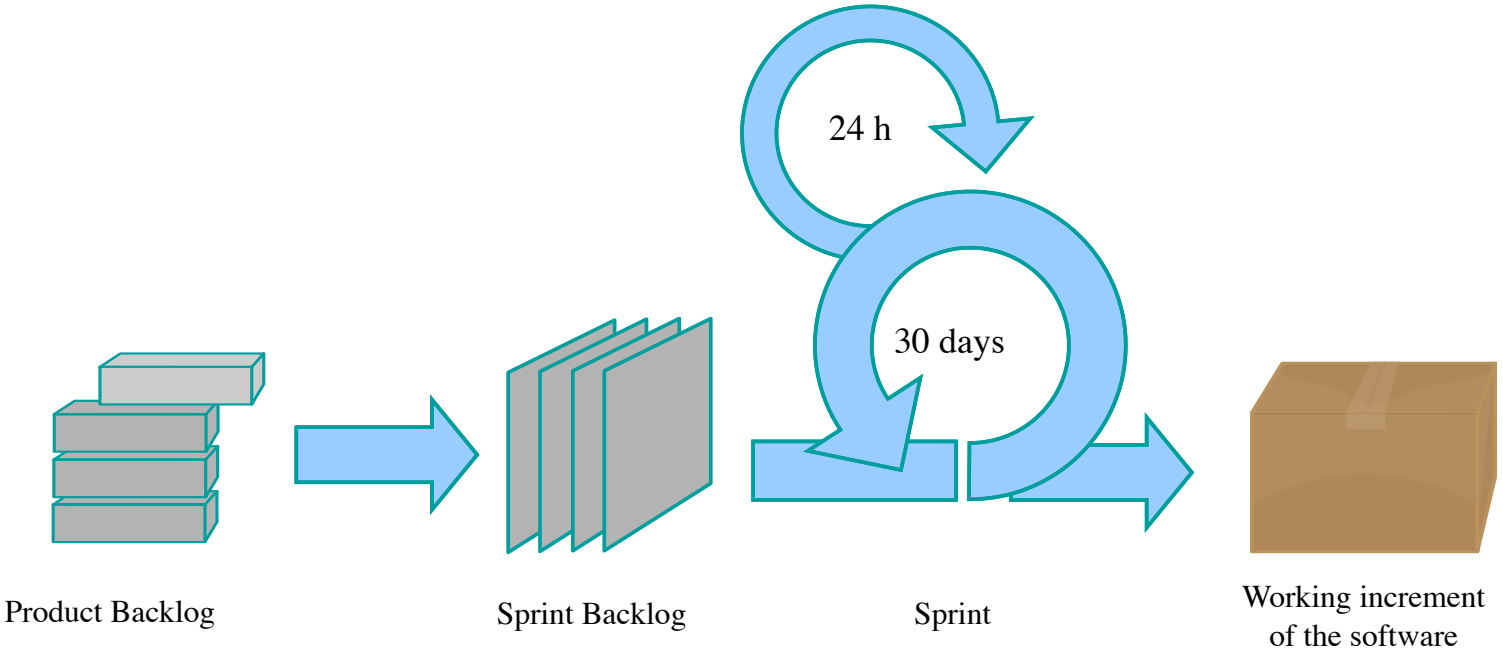
eXtreme Programming (XP)

- Kent Beck 1999
- 12 Practices



Kent Beck, Extreme Programming 1st ed.

Scrum



Wikipedia

- Robert Martin (Uncle Bob) about ”The Land that Scrum Forgot”
  `http://www.youtube.com/watch?v=hG4LH6P8Syk`
  → History about agile methods, the agile manifesto, and Scrum and its relationshop to XP

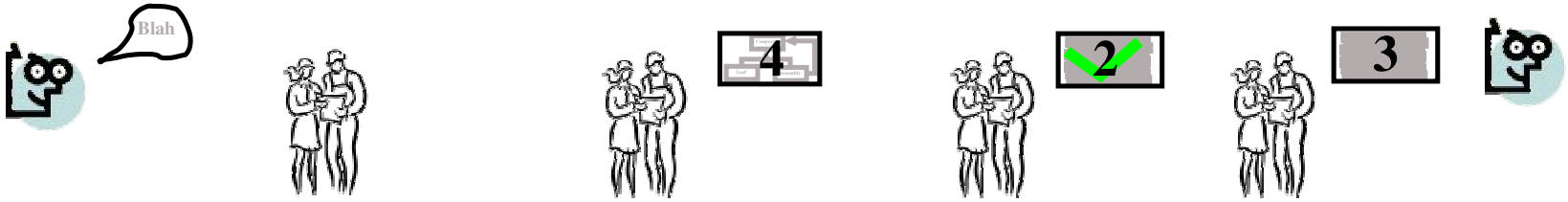
Lean Software Development

- Principles of Lean Production: coming from Toyota
  - Value for the customer
  - Reduce the amount of waste in the production process
  - Generate flow
- Waste: resources used which do not produce value for the customer
  - time needed to fix bugs
    - → Test/behaviour-driven development
  - time to change the system because it does not fit the customers requirements
    - → Customer on-site; build a system get feedback
  - time for building infrastructure for future features
    - → YAGNI (You Ain't Gona Need It)
  - ...

Generating flow using Pull and Kanban

WIP = Work in Progress Limit

Work Item	A Queue	A WIP 3	D Queue	D WIP 3	T Queue	T WIP 3	I Queue	I WIP 3	Done
6, 5, 7, 8, 10, 9				4		2		3	1



Flow through Pull with Kanban

- Process controlling: local rules
- Load balancing: Kanban cards and Work in Progress (WIP) limits
- Integration in other processes

Figure from David Anderson `www.agilemanagement.net`

Online Kanban Tool: Trello

- `www.trello.com`: Electronic Kanban board useful for your project
- Example `https://trello.com/b/4wddd1zf/kanban-workflow`

→ use recommended to keep track of user stories/features/tasks